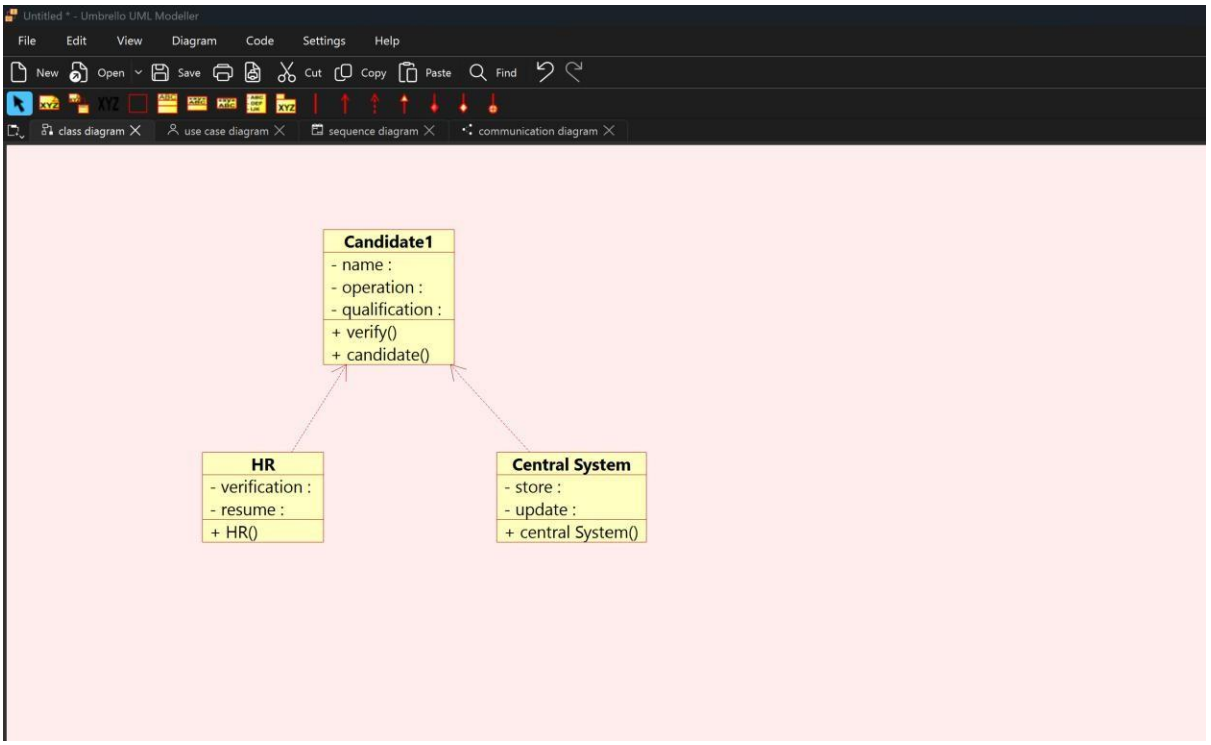
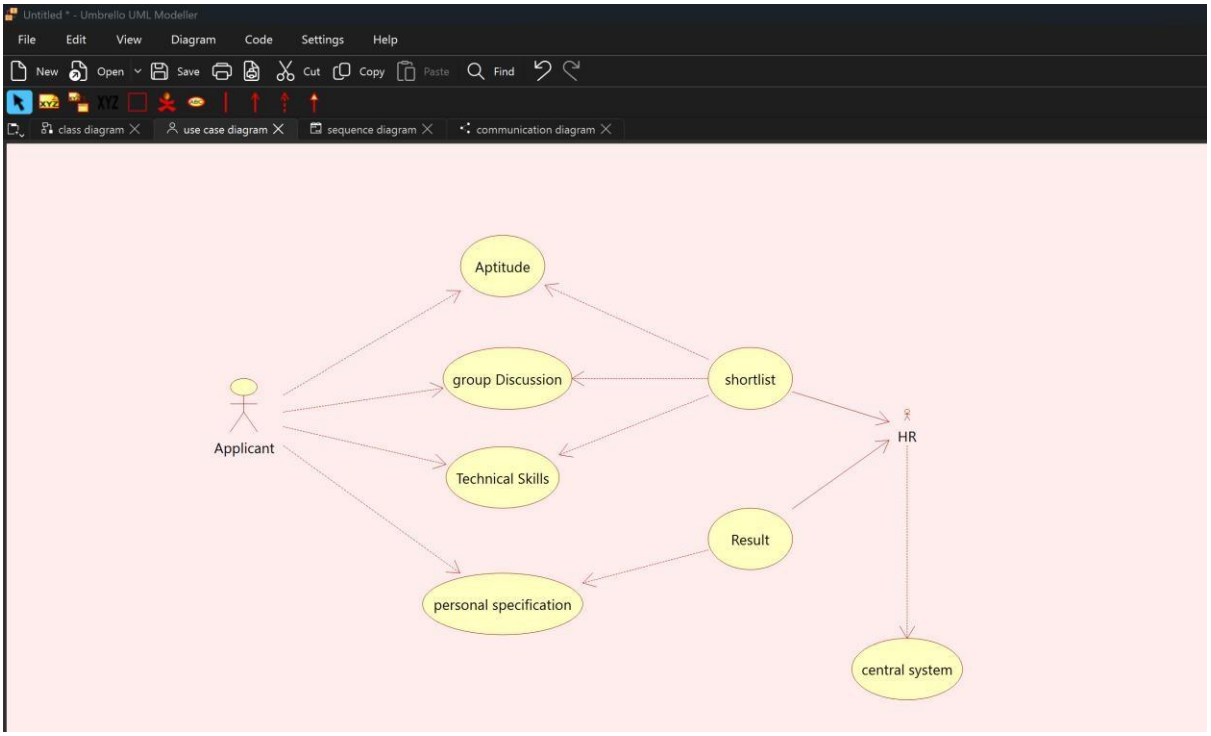
Ex No: 10 – RECRUITMENT SYSTEM

NAME: Arava Raja (192365033)

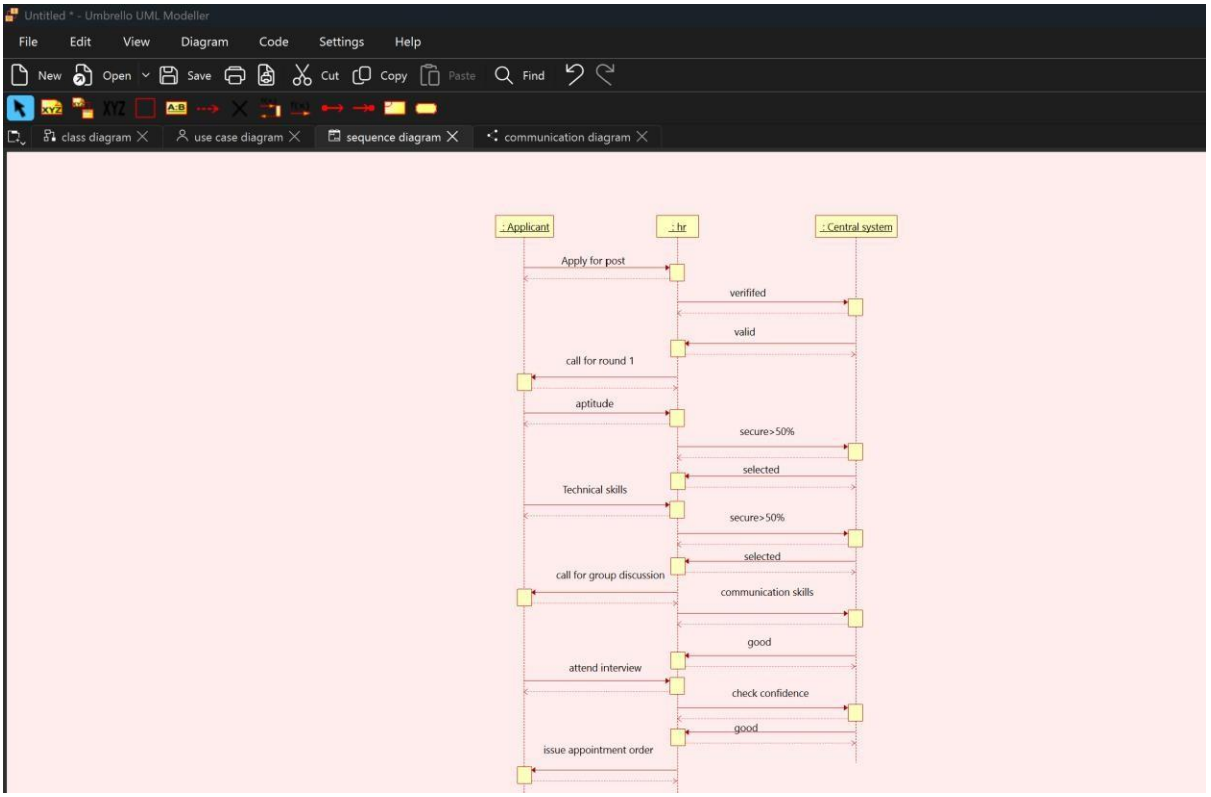
CLASS DIAGRAM:



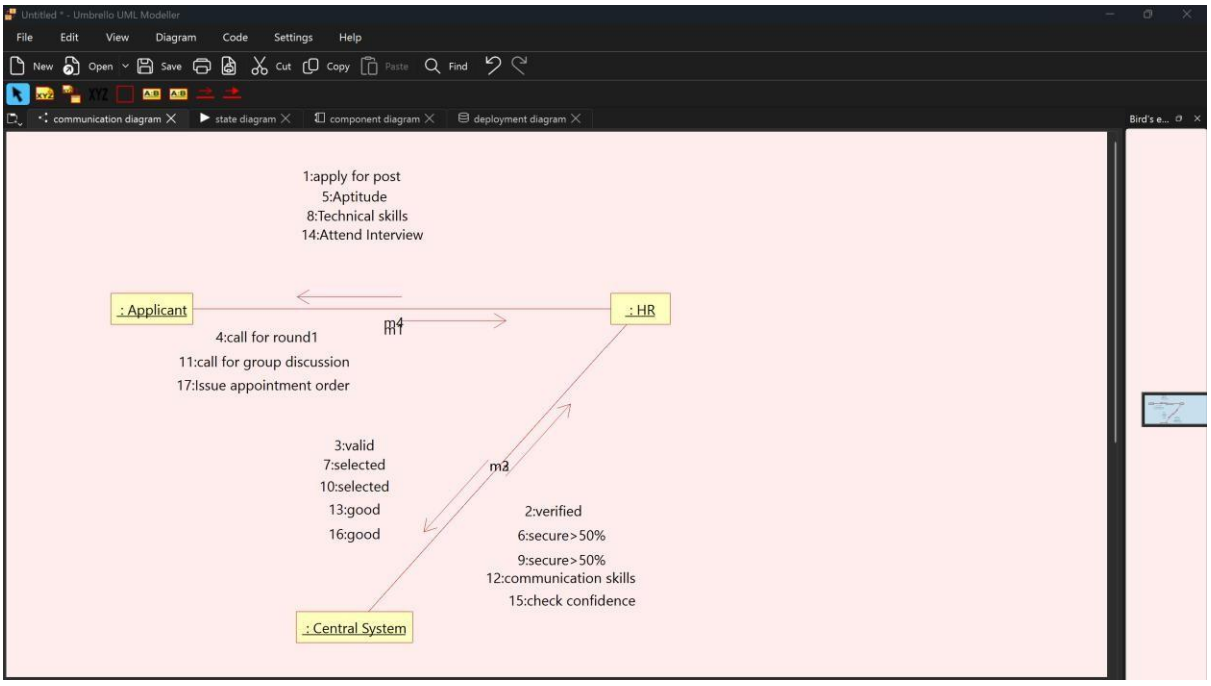
USE CASE DIAGRAM:



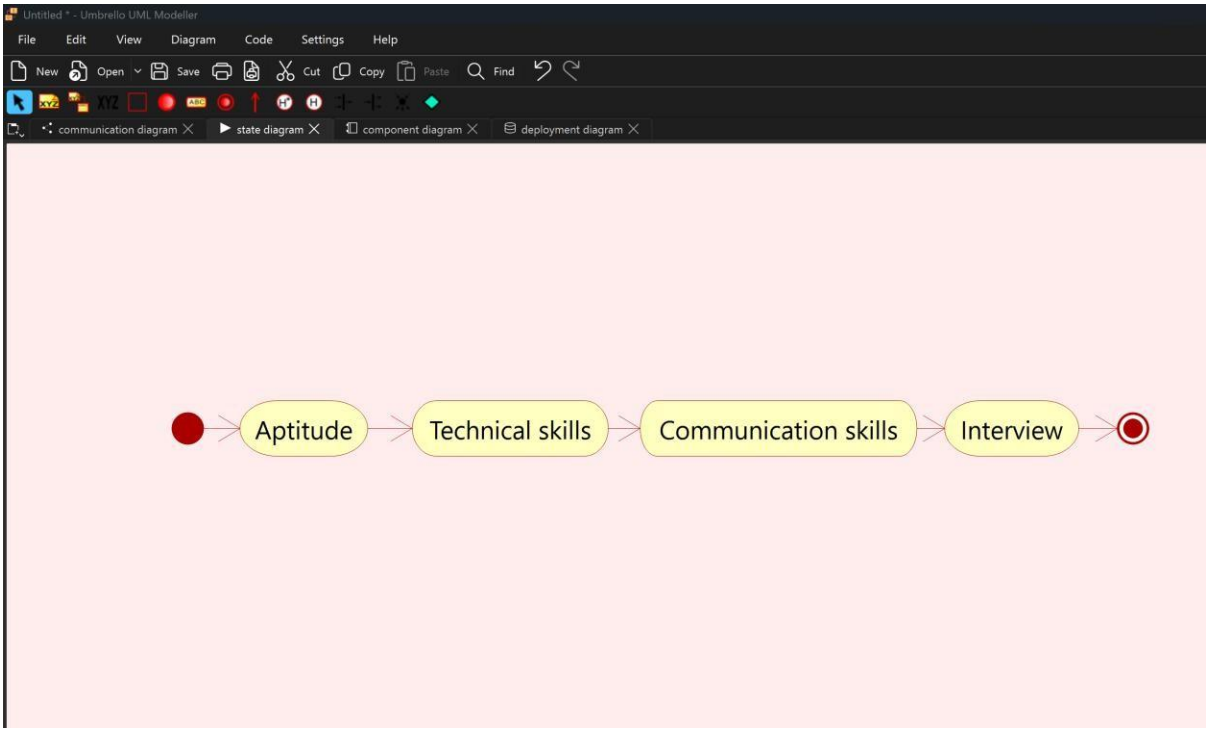
SEQUENCE DIAGRAM:



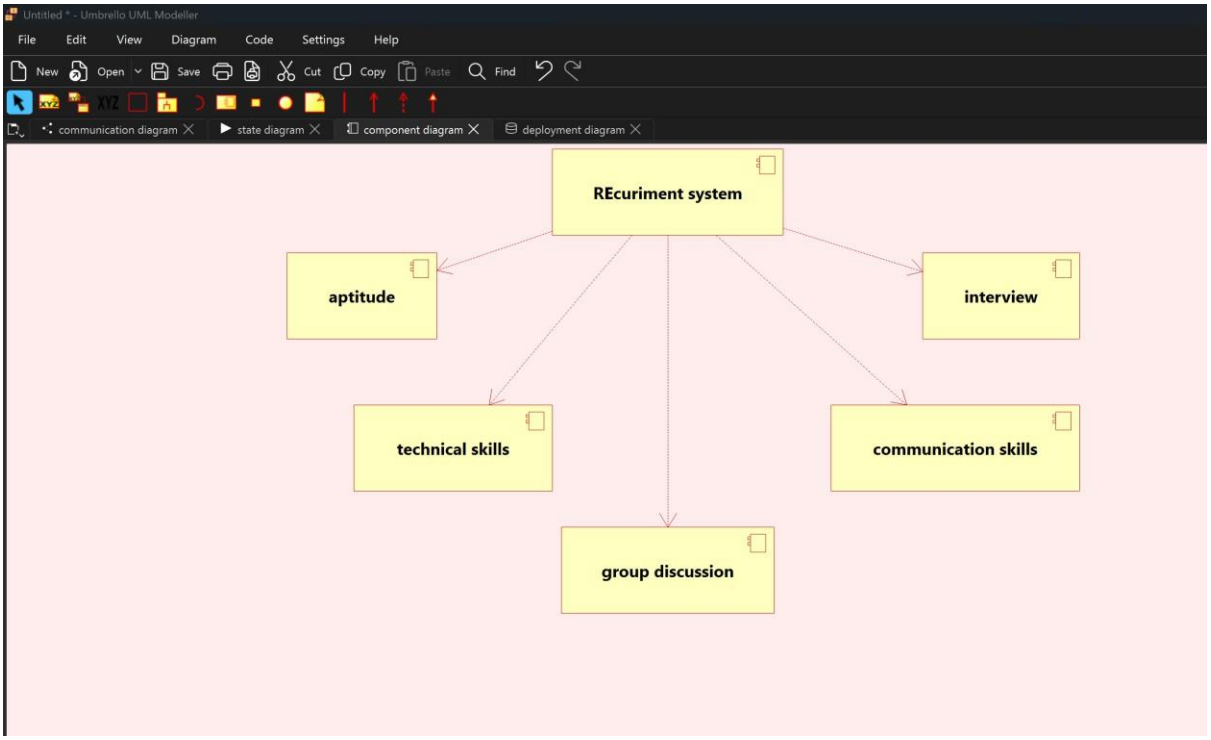
COLLABORATION DIAGRAM:



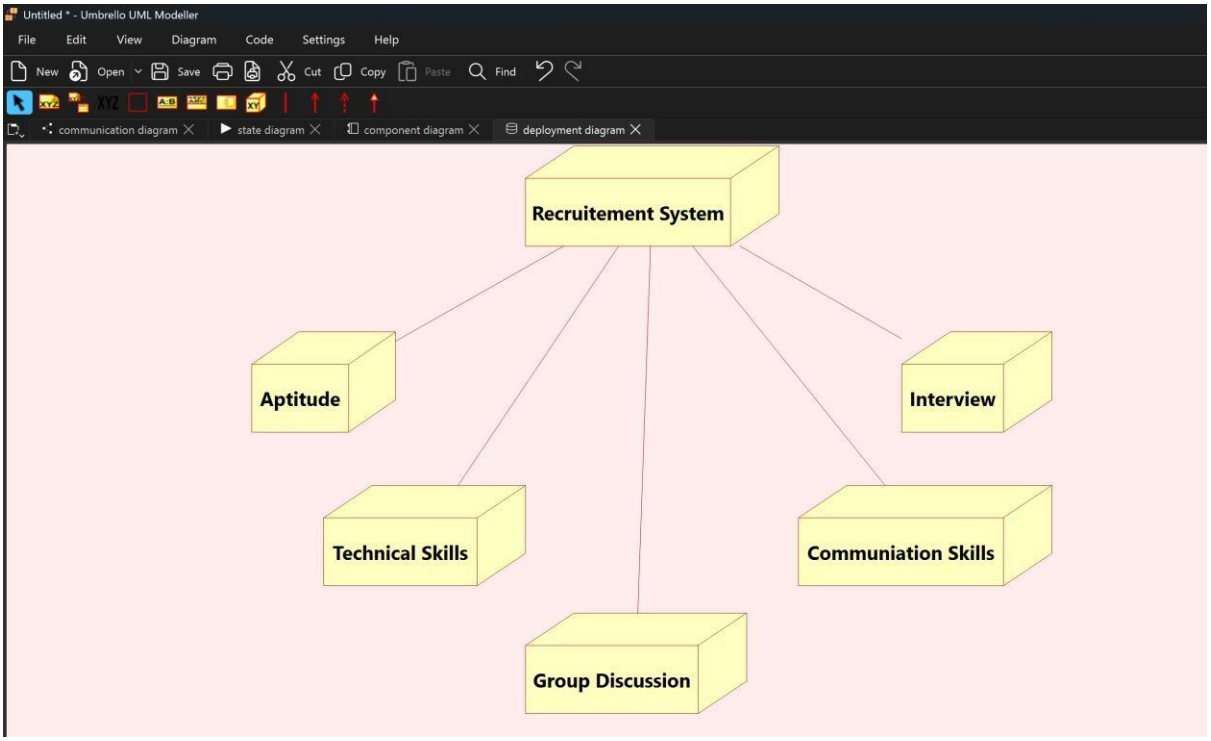
STATE DIAGRAM



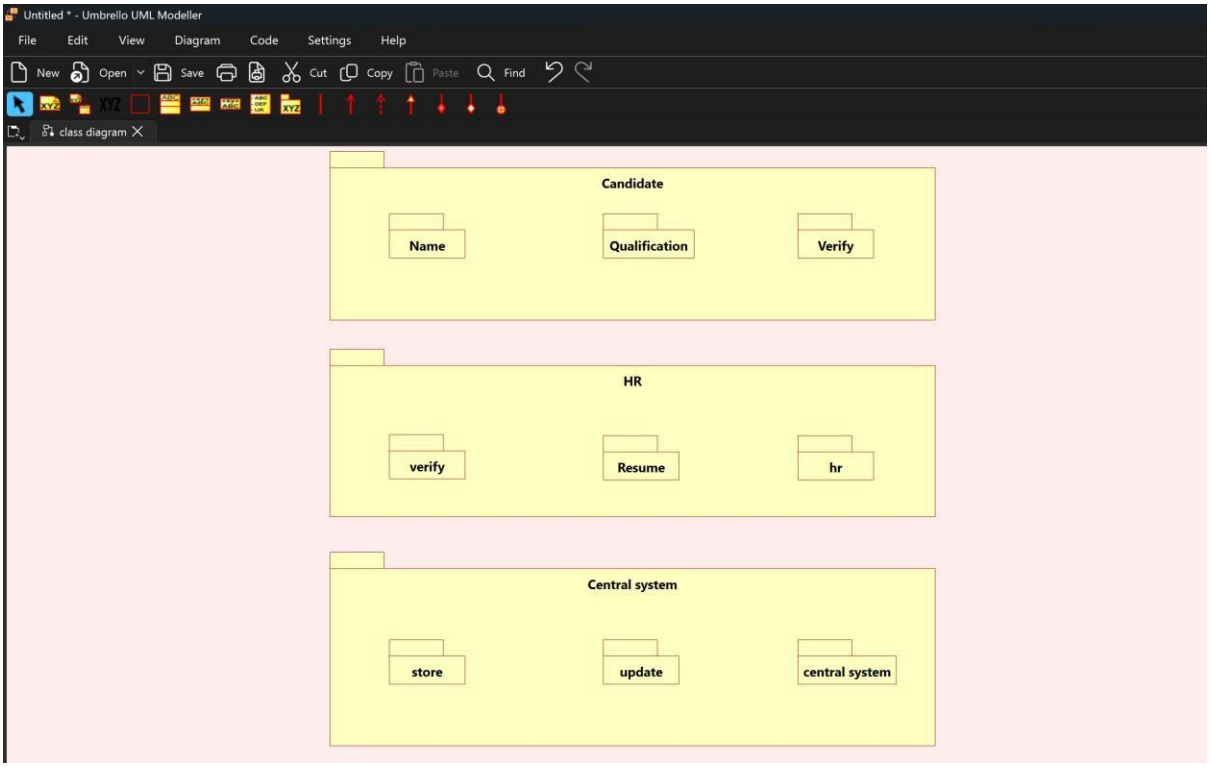
COMPONENT DIAGRAM;



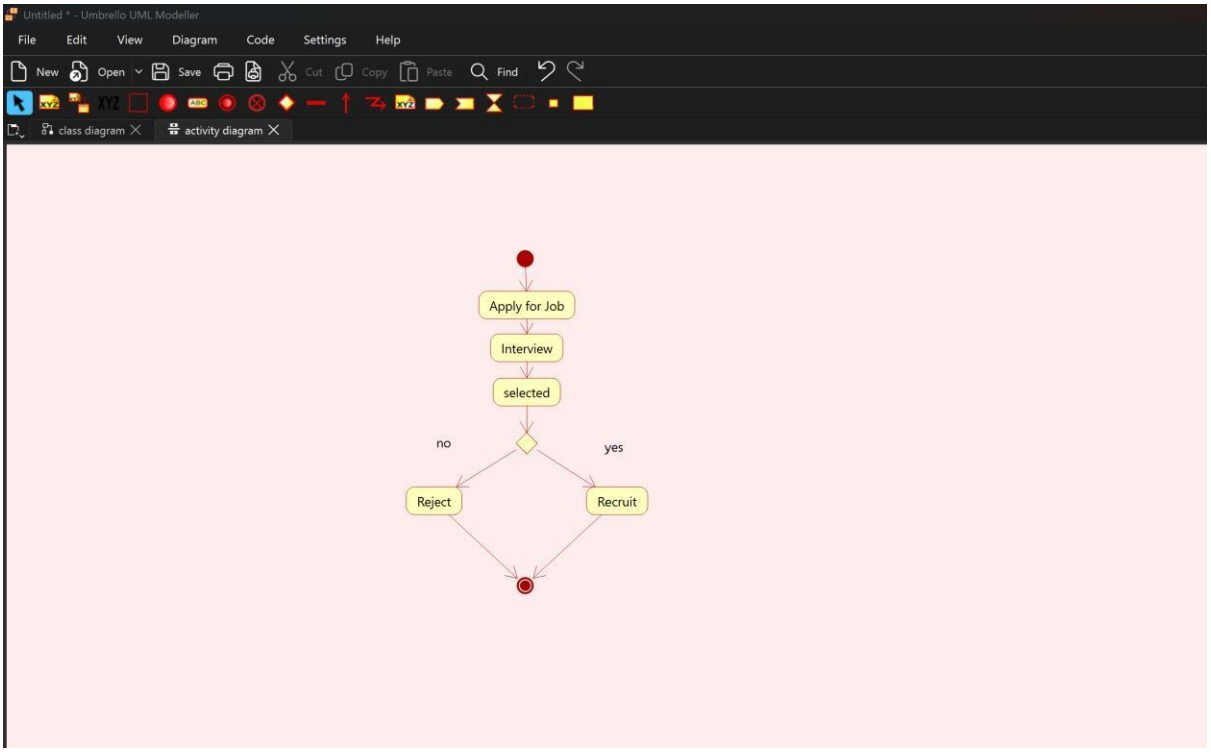
DEPLOYMENT DIAGRAM:



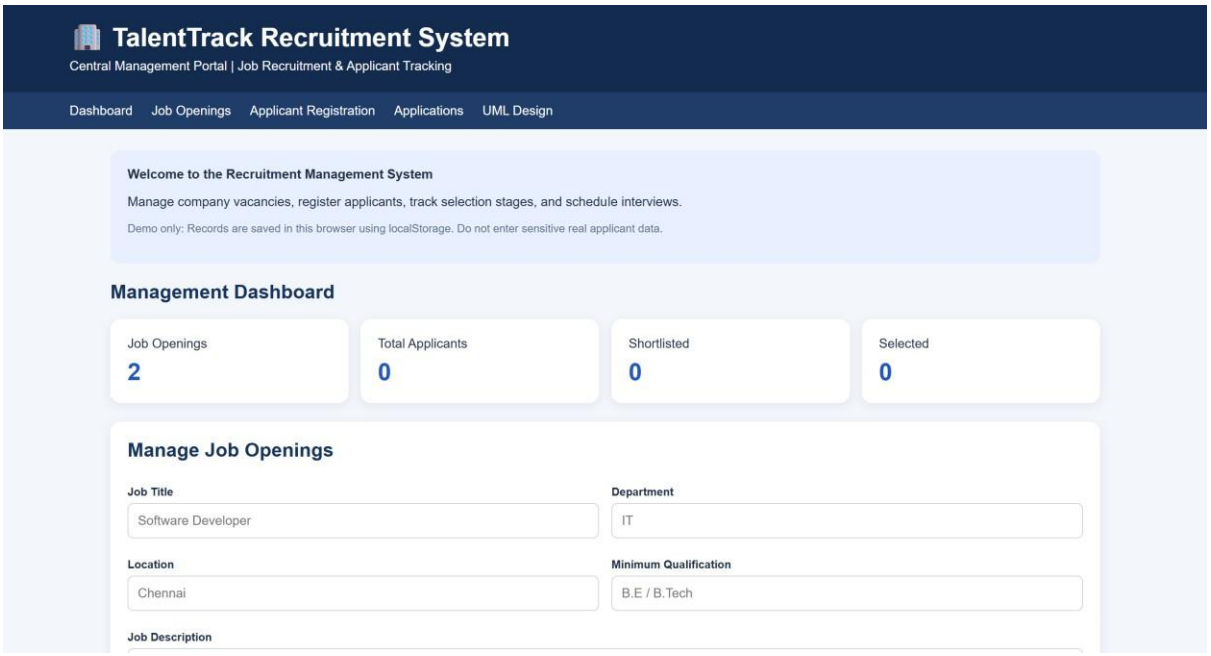
PACKAGE DIAGRAM



ACTIVITY DIAGRAM:



WEBPAGE FOR RECRUITMENT SYSTEM



Manage Job Openings

Job Title

Software Developer

Department

IT

Location

Chennai

Minimum Qualification

B.E / B.Tech

Job Description

Add Job Opening

Current Vacancies

Job ID	Title	Department	Location	Qualification	Action
JOB101	Software Developer	IT	Chennai	B.E / B.Tech	Delete
JOB102	QA Tester	Quality Assurance	Bengaluru	Any Computer Science Degree	Delete

JOB102	QA Tester	Quality Assurance	Bengaluru	Any Computer Science Degree	Delete

Applicant Registration

Full Name

Email

Phone

Qualification

Years of Experience

Applying For

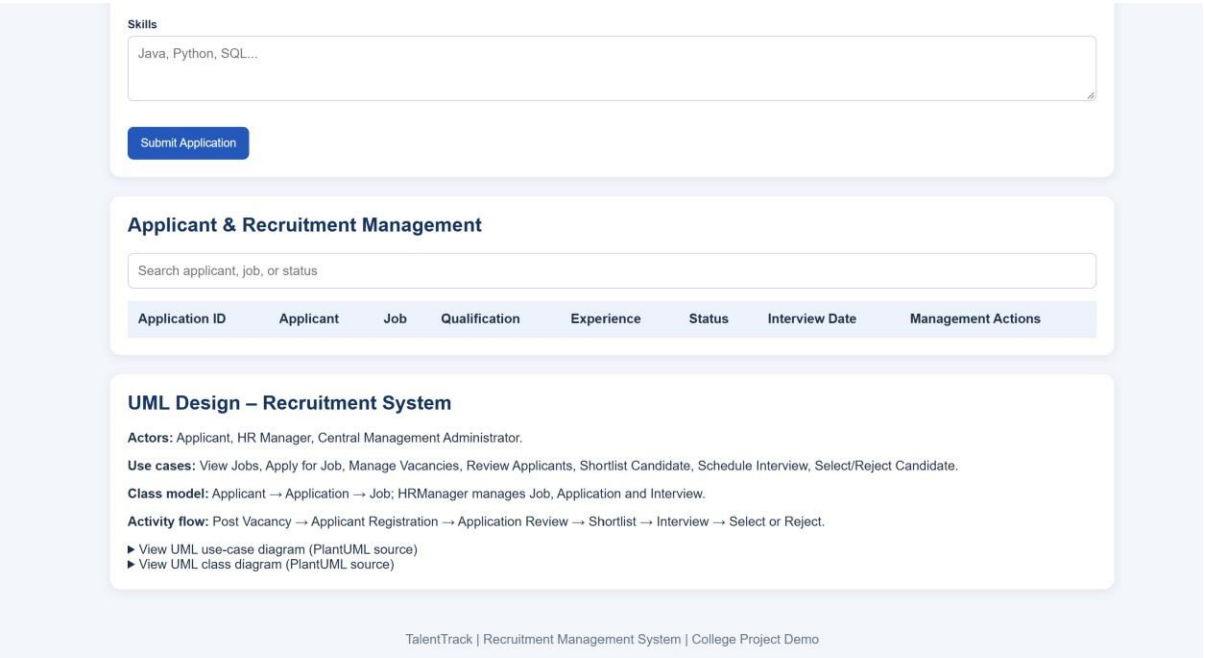
Select Job

Skills

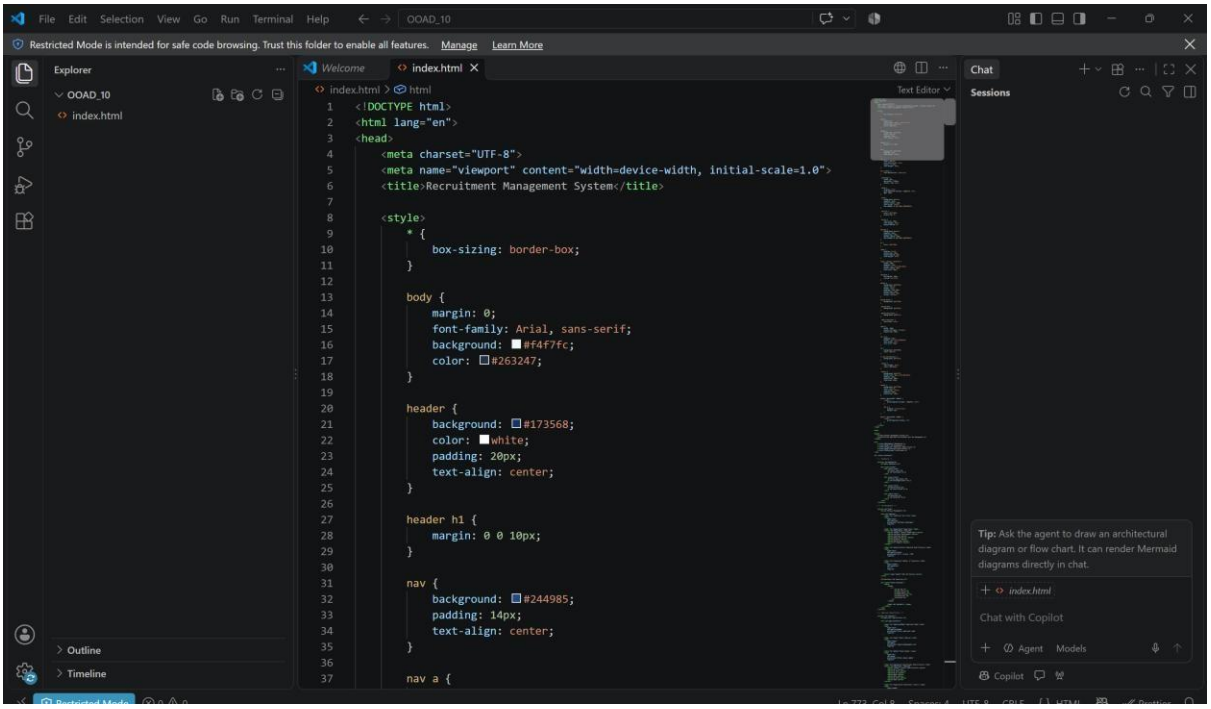
Java, Python, SQL...

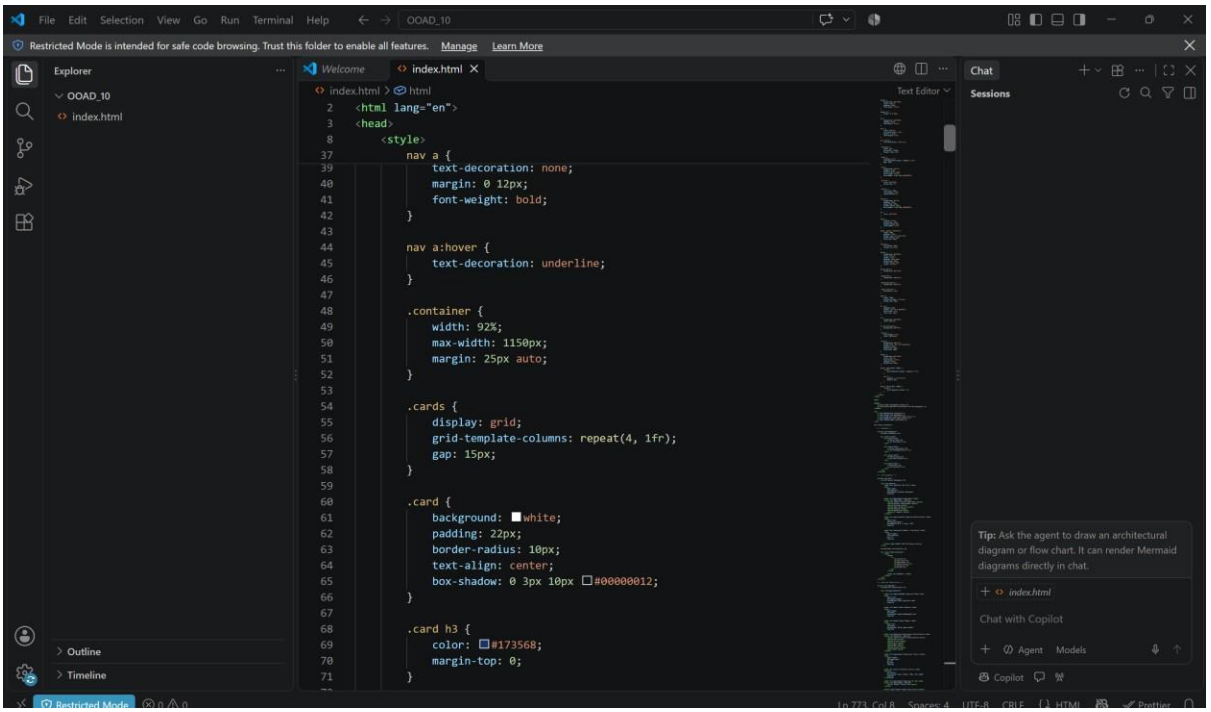
Submit Application

Applicant & Recruitment Management


Skills
Java, Python, SQL...
Submit Application
Applicant & Recruitment Management
Search applicant, job, or status
Application ID
Applicant
Job
Qualification
Experience
Status
Interview Date
Management Actions
UML Design – Recruitment System
Actors: Applicant, HR Manager, Central Management Administrator.
Use cases: View Jobs, Apply for Job, Manage Vacancies, Review Applicants, Shortlist Candidate, Schedule Interview, Select/Reject Candidate.
Class model: Applicant → Application → Job; HRManager manages Job, Application and Interview.
Activity flow: Post Vacancy → Applicant Registration → Application Review → Shortlist → Interview → Select or Reject.
▶ View UML use-case diagram (PlantUML source)
▶ View UML class diagram (PlantUML source)
TalentTrack | Recruitment Management System | College Project Demo

HTML CODE:


File Edit Selection View Go Run Terminal Help
OOAD_10
Restricted Mode is intended for safe code browsing. Trust this folder to enable all features. Manage Learn More
Explorer
OOAD_10
index.html
Welcome
index.html
Chat
Sessions
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Recruitment Management System</title>
<style>
* {
box-sizing: border-box;
}
body {
margin: 0;
font-family: Arial, sans-serif;
background: #f4f7fc;
color: #263247;
}
header {
background: #173568;
color: white;
padding: 20px;
text-align: center;
}
header h1 {
margin: 0 0 10px;
}
nav {
background: #244985;
padding: 14px;
text-align: center;
}
nav a {
Tip: Ask the agent to draw an architectural diagram or flow chart. It can render Mermaid diagrams directly in chat.
Chat with Copilot
Agent
Models
Outline
Timeline




```
<html lang="en">
<head>
    <style>
        nav a {
            text-decoration: none;
            margin: 0 12px;
            font-weight: bold;
        }

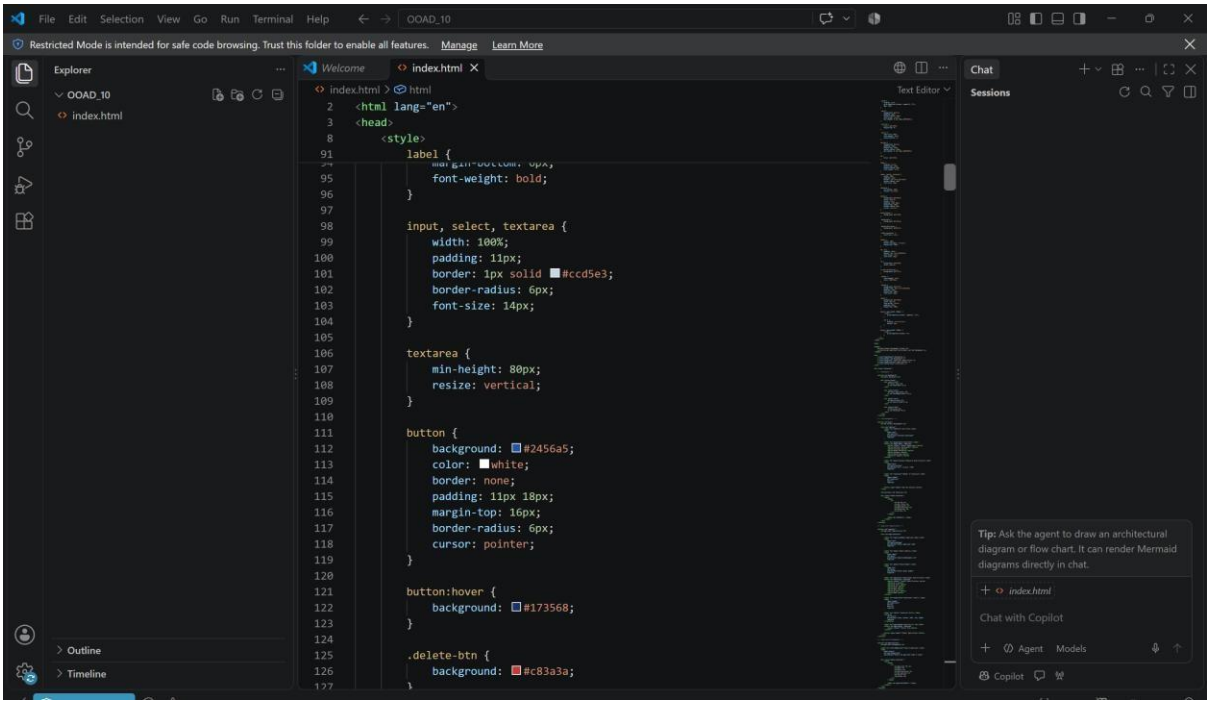
        nav a:hover {
            text-decoration: underline;
        }

        .container {
            width: 92%;
            max-width: 1150px;
            margin: 25px auto;
        }

        .cards {
            display: grid;
            grid-template-columns: repeat(4, 1fr);
            gap: 15px;
        }

        .card {
            background: white;
            padding: 22px;
            border-radius: 10px;
            text-align: center;
            box-shadow: 0 3px 10px #00000012;
        }

        .card h3 {
            color: #173568;
            margin-top: 0;
        }
```




```
<html lang="en">
<head>
    <style>
        label {
            font-weight: bold;
        }

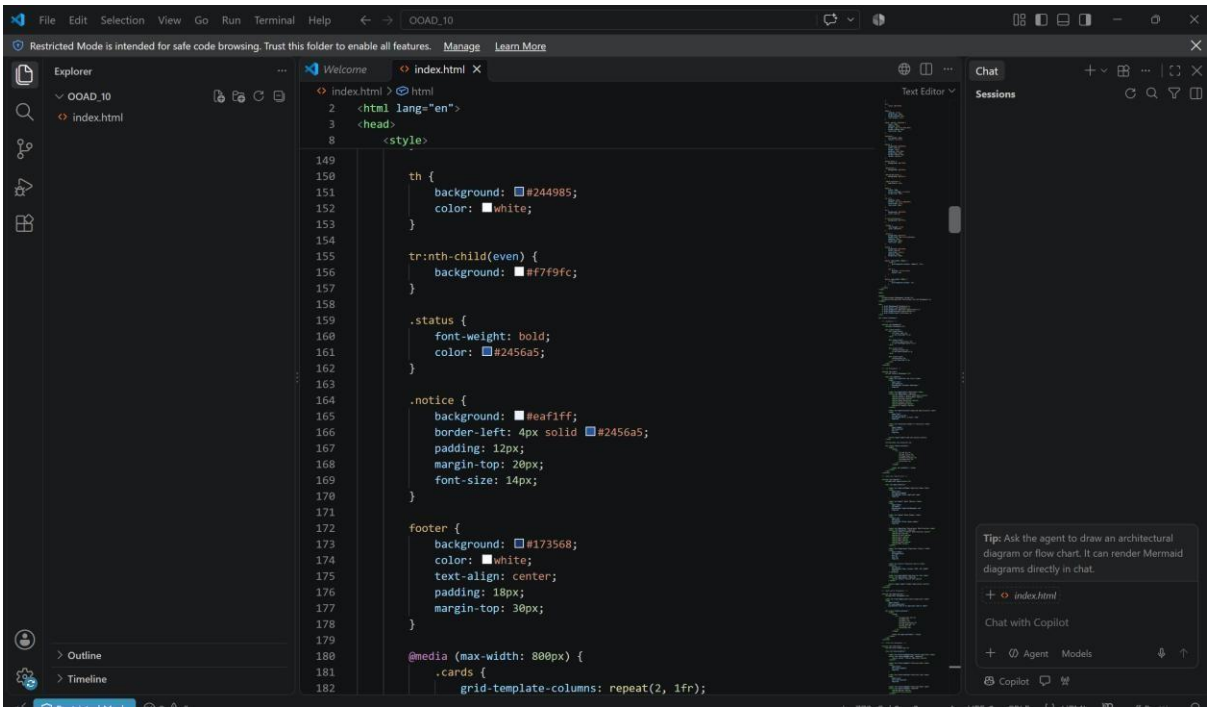
        input, select, textarea {
            width: 100%;
            padding: 11px;
            border: 1px solid #ccd5e3;
            border-radius: 6px;
            font-size: 14px;
        }

        textarea {
            min-height: 80px;
            resize: vertical;
        }

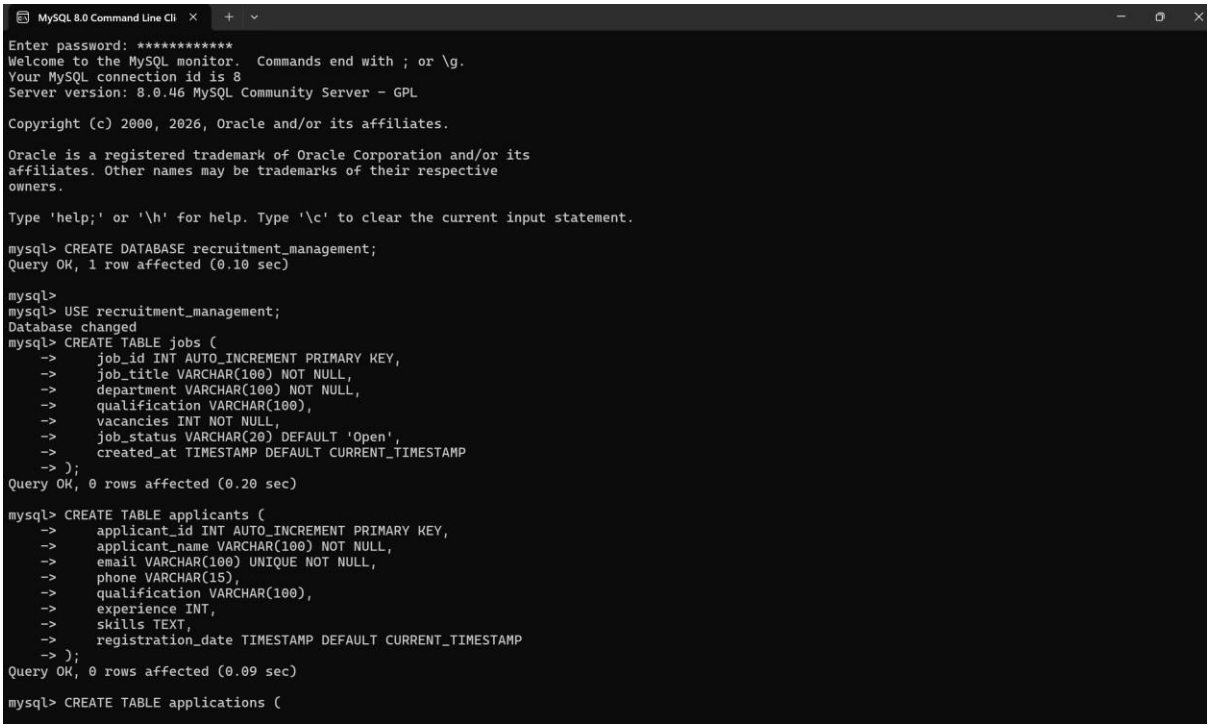
        button {
            background: #2456a5;
            color: white;
            border: none;
            padding: 11px 18px;
            margin-top: 16px;
            border-radius: 6px;
            cursor: pointer;
        }

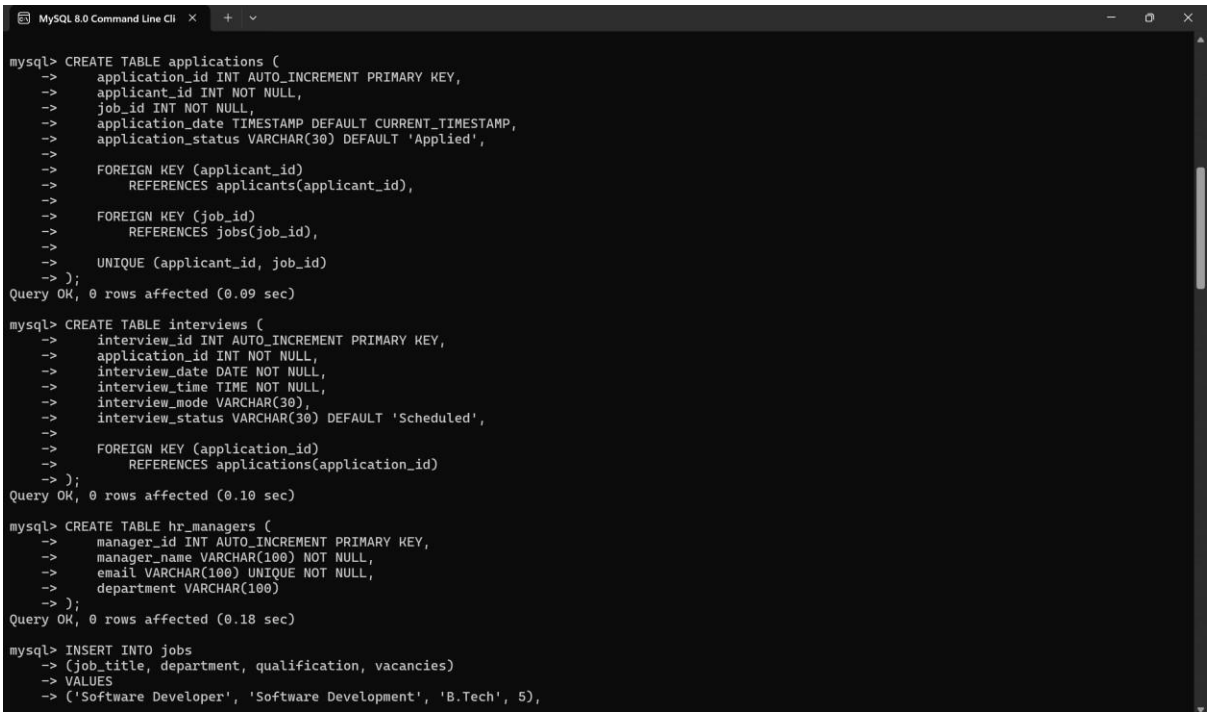
        button:hover {
            background: #173568;
        }

        .delete-btn {
            background: #c83a3a;
```



MYSQL DATABASE:





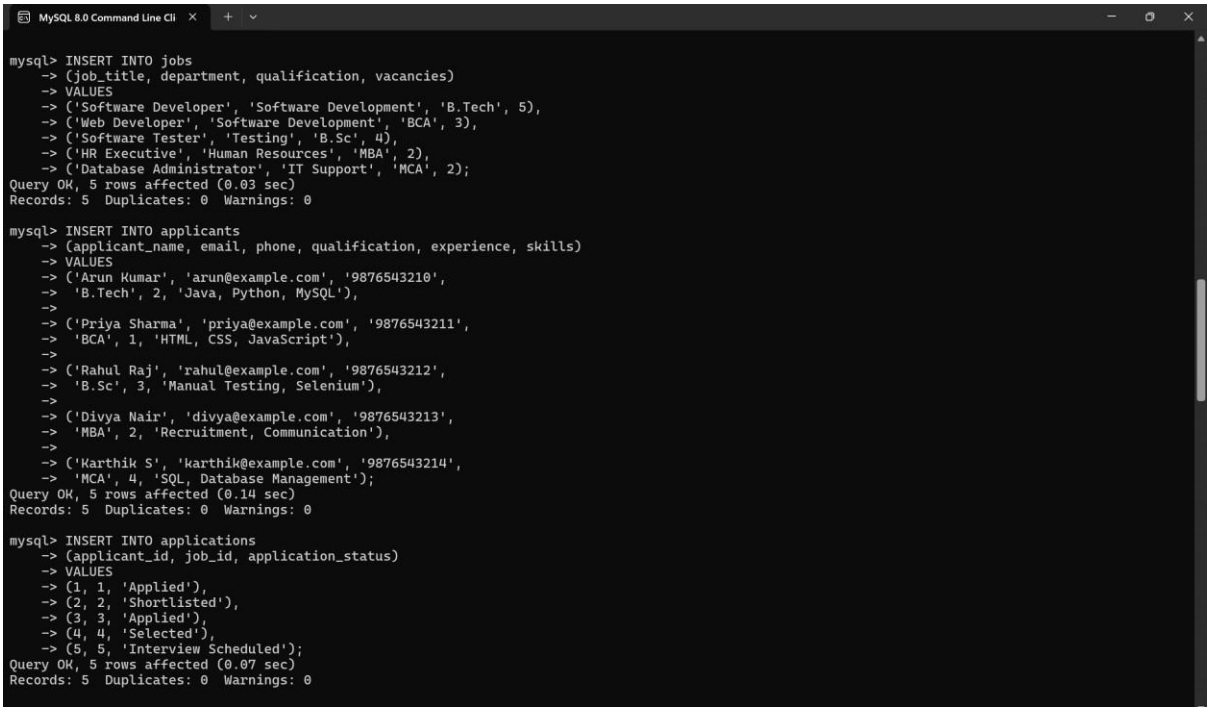
```
MySQL 8.0 Command Line Cli

mysql> CREATE TABLE applications (
    ->     application_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     applicant_id INT NOT NULL,
    ->     job_id INT NOT NULL,
    ->     application_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    ->     application_status VARCHAR(30) DEFAULT 'Applied',
    ->
    ->     FOREIGN KEY (applicant_id)
    ->         REFERENCES applicants(applicant_id),
    ->
    ->     FOREIGN KEY (job_id)
    ->         REFERENCES jobs(job_id),
    ->
    ->     UNIQUE (applicant_id, job_id)
    -> );
Query OK, 0 rows affected (0.09 sec)

mysql> CREATE TABLE interviews (
    ->     interview_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     application_id INT NOT NULL,
    ->     interview_date DATE NOT NULL,
    ->     interview_time TIME NOT NULL,
    ->     interview_mode VARCHAR(30),
    ->     interview_status VARCHAR(30) DEFAULT 'Scheduled',
    ->
    ->     FOREIGN KEY (application_id)
    ->         REFERENCES applications(application_id)
    -> );
Query OK, 0 rows affected (0.10 sec)

mysql> CREATE TABLE hr_managers (
    ->     manager_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     manager_name VARCHAR(100) NOT NULL,
    ->     email VARCHAR(100) UNIQUE NOT NULL,
    ->     department VARCHAR(100)
    -> );
Query OK, 0 rows affected (0.18 sec)

mysql> INSERT INTO jobs
    -> (job_title, department, qualification, vacancies)
    -> VALUES
    -> ('Software Developer', 'Software Development', 'B.Tech', 5),
```

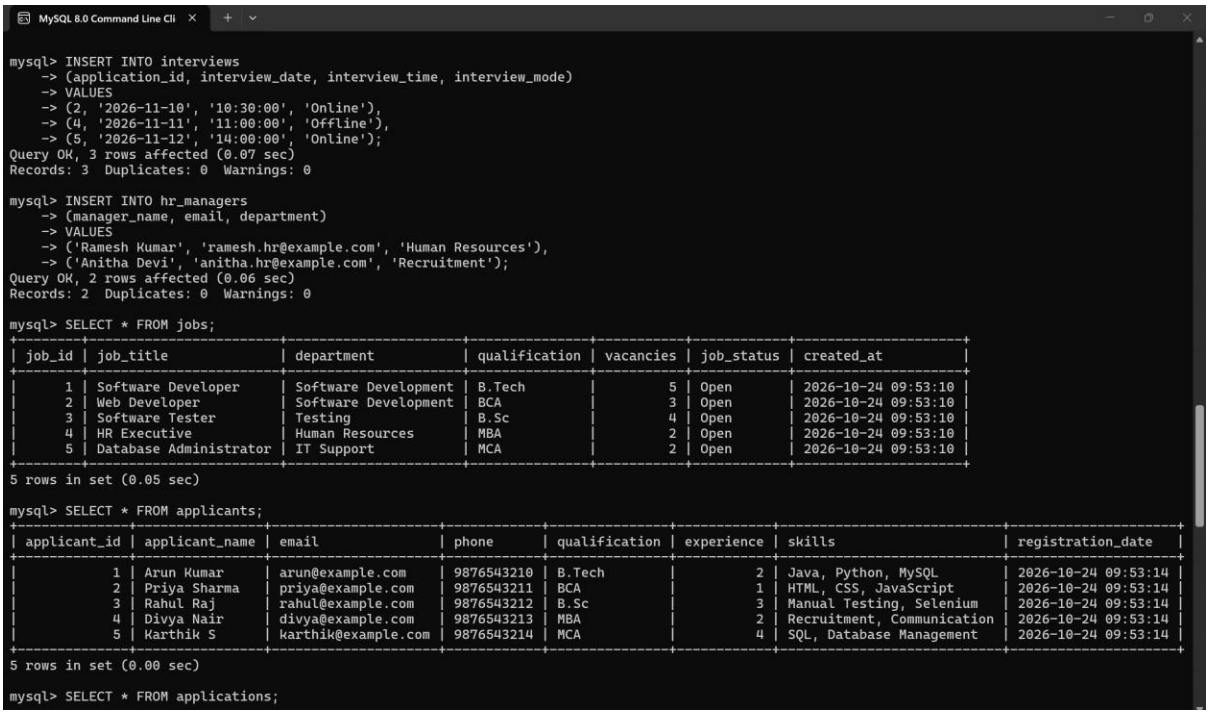


```
MySQL 8.0 Command Line Cli

mysql> INSERT INTO jobs
    -> (job_title, department, qualification, vacancies)
    -> VALUES
    -> ('Software Developer', 'Software Development', 'B.Tech', 5),
    -> ('Web Developer', 'Software Development', 'BCA', 3),
    -> ('Software Tester', 'Testing', 'B.Sc', 4),
    -> ('HR Executive', 'Human Resources', 'MBA', 2),
    -> ('Database Administrator', 'IT Support', 'MCA', 2);
Query OK, 5 rows affected (0.03 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> INSERT INTO applicants
    -> (applicant_name, email, phone, qualification, experience, skills)
    -> VALUES
    -> ('Arun Kumar', 'arun@example.com', '9876543210',
    ->  'B.Tech', 2, 'Java, Python, MySQL'),
    ->
    -> ('Priya Sharma', 'priya@example.com', '9876543211',
    ->  'BCA', 1, 'HTML, CSS, JavaScript'),
    ->
    -> ('Rahul Raj', 'rahul@example.com', '9876543212',
    ->  'B.Sc', 3, 'Manual Testing, Selenium'),
    ->
    -> ('Divya Nair', 'divya@example.com', '9876543213',
    ->  'MBA', 2, 'Recruitment, Communication'),
    ->
    -> ('Karthik S', 'karthik@example.com', '9876543214',
    ->  'MCA', 4, 'SQL, Database Management');
Query OK, 5 rows affected (0.14 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> INSERT INTO applications
    -> (applicant_id, job_id, application_status)
    -> VALUES
    -> (1, 1, 'Applied'),
    -> (2, 2, 'Shortlisted'),
    -> (3, 3, 'Applied'),
    -> (4, 4, 'Selected'),
    -> (5, 5, 'Interview Scheduled');
Query OK, 5 rows affected (0.07 sec)
Records: 5  Duplicates: 0  Warnings: 0
```



```
MySQL 8.0 Command Line Cli

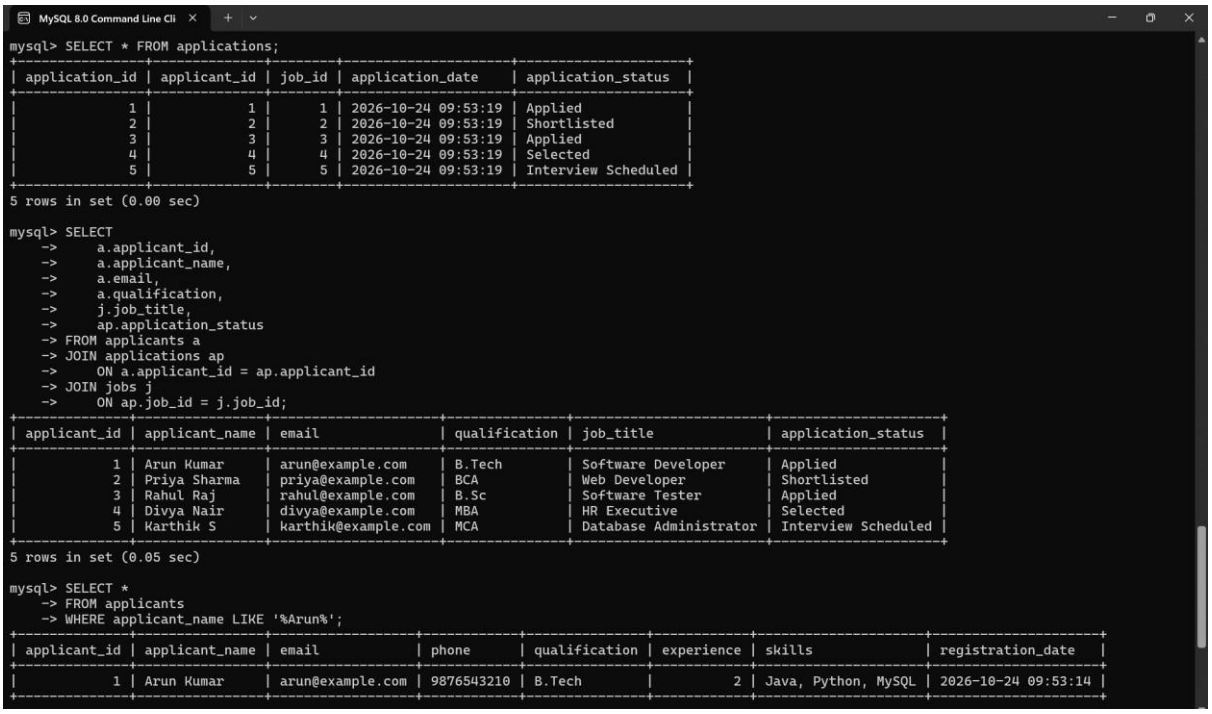
mysql> INSERT INTO interviews
    -> (application_id, interview_date, interview_time, interview_mode)
    -> VALUES
    -> (2, '2026-11-10', '10:30:00', 'Online'),
    -> (4, '2026-11-11', '11:00:00', 'Offline'),
    -> (5, '2026-11-12', '14:00:00', 'Online');
Query OK, 3 rows affected (0.07 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO hr_managers
    -> (manager_name, email, department)
    -> VALUES
    -> ('Ramesh Kumar', 'ramesh.hr@example.com', 'Human Resources'),
    -> ('Anitha Devi', 'anitha.hr@example.com', 'Recruitment');
Query OK, 2 rows affected (0.06 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM jobs;
+--------+-------------------------+----------------------+---------------+-----------+------------+---------------------+
| job_id | job_title               | department           | qualification | vacancies | job_status | created_at          |
+--------+-------------------------+----------------------+---------------+-----------+------------+---------------------+
|      1 | Software Developer      | Software Development | B.Tech        |         5 | Open       | 2026-10-24 09:53:10 |
|      2 | Web Developer           | Software Development | BCA           |         3 | Open       | 2026-10-24 09:53:10 |
|      3 | Software Tester         | Testing              | B.Sc          |         4 | Open       | 2026-10-24 09:53:10 |
|      4 | HR Executive            | Human Resources      | MBA           |         2 | Open       | 2026-10-24 09:53:10 |
|      5 | Database Administrator | IT Support           | MCA           |         2 | Open       | 2026-10-24 09:53:10 |
+--------+-------------------------+----------------------+---------------+-----------+------------+---------------------+
5 rows in set (0.05 sec)

mysql> SELECT * FROM applicants;
+--------------+----------------+-------------------+------------+---------------+------------+---------------------------+---------------------+
| applicant_id | applicant_name | email             | phone      | qualification | experience | skills                    | registration_date   |
+--------------+----------------+-------------------+------------+---------------+------------+---------------------------+---------------------+
|            1 | Arun Kumar     | arun@example.com  | 9876543210 | B.Tech        |          2 | Java, Python, MySQL       | 2026-10-24 09:53:14 |
|            2 | Priya Sharma   | priya@example.com | 9876543211 | BCA           |          1 | HTML, CSS, JavaScript     | 2026-10-24 09:53:14 |
|            3 | Rahul Raj      | rahul@example.com | 9876543212 | B.Sc          |          3 | Manual Testing, Selenium  | 2026-10-24 09:53:14 |
|            4 | Divya Nair     | divya@example.com | 9876543213 | MBA           |          2 | Recruitment, Communication | 2026-10-24 09:53:14 |
|            5 | Karthik S      | karthik@example.com | 9876543214 | MCA           |          4 | SQL, Database Management  | 2026-10-24 09:53:14 |
+--------------+----------------+-------------------+------------+---------------+------------+---------------------------+---------------------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM applications;
```

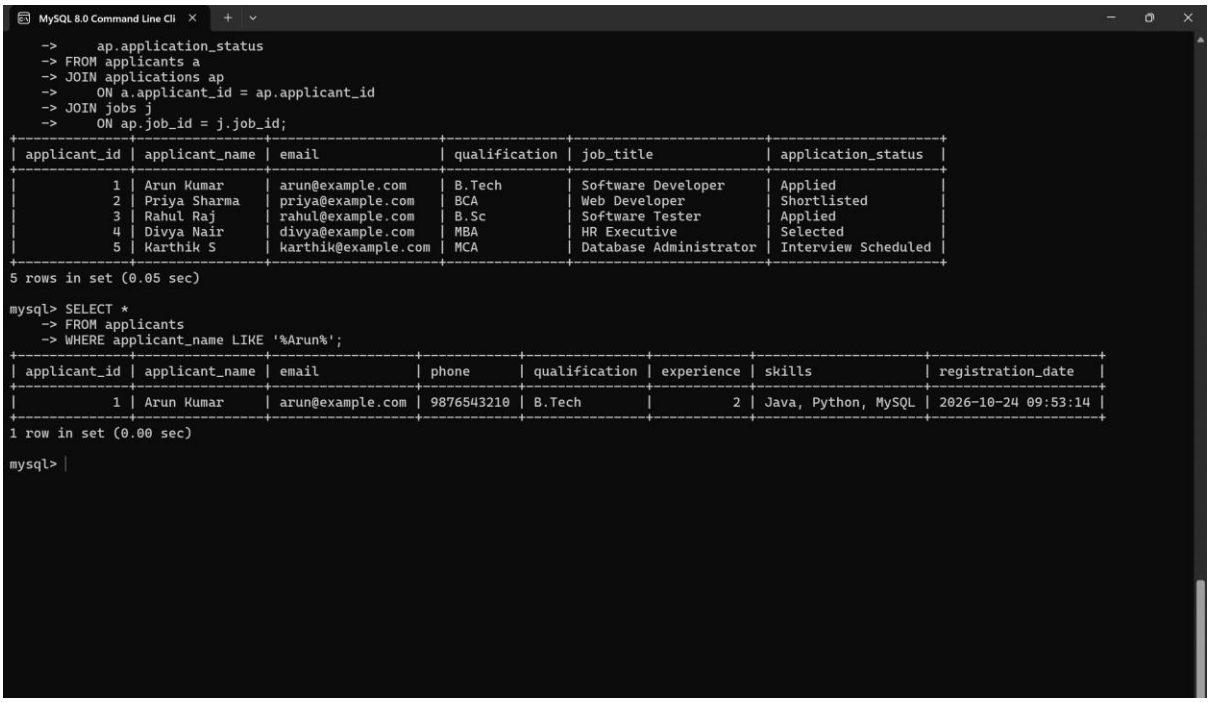


```
MySQL 8.0 Command Line Cli

mysql> SELECT * FROM applications;
+----------------+--------------+--------+---------------------+----------------------+
| application_id | applicant_id | job_id | application_date    | application_status   |
+----------------+--------------+--------+---------------------+----------------------+
|              1 |            1 |      1 | 2026-10-24 09:53:19 | Applied              |
|              2 |            2 |      2 | 2026-10-24 09:53:19 | Shortlisted          |
|              3 |            3 |      3 | 2026-10-24 09:53:19 | Applied              |
|              4 |            4 |      4 | 2026-10-24 09:53:19 | Selected             |
|              5 |            5 |      5 | 2026-10-24 09:53:19 | Interview Scheduled  |
+----------------+--------------+--------+---------------------+----------------------+
5 rows in set (0.00 sec)

mysql> SELECT
    ->     a.applicant_id,
    ->     a.applicant_name,
    ->     a.email,
    ->     a.qualification,
    ->     j.job_title,
    ->     ap.application_status
    -> FROM applicants a
    -> JOIN applications ap
    ->     ON a.applicant_id = ap.applicant_id
    -> JOIN jobs j
    ->     ON ap.job_id = j.job_id;
+--------------+----------------+-------------------+---------------+-------------------------+----------------------+
| applicant_id | applicant_name | email             | qualification | job_title               | application_status   |
+--------------+----------------+-------------------+---------------+-------------------------+----------------------+
|            1 | Arun Kumar     | arun@example.com  | B.Tech        | Software Developer      | Applied              |
|            2 | Priya Sharma   | priya@example.com | BCA           | Web Developer           | Shortlisted          |
|            3 | Rahul Raj      | rahul@example.com | B.Sc          | Software Tester         | Applied              |
|            4 | Divya Nair     | divya@example.com | MBA           | HR Executive            | Selected             |
|            5 | Karthik S      | karthik@example.com | MCA           | Database Administrator | Interview Scheduled  |
+--------------+----------------+-------------------+---------------+-------------------------+----------------------+
5 rows in set (0.05 sec)

mysql> SELECT *
    -> FROM applicants
    -> WHERE applicant_name LIKE '%Arun%';
+--------------+----------------+-----------------+------------+---------------+------------+--------------------+---------------------+
| applicant_id | applicant_name | email           | phone      | qualification | experience | skills             | registration_date   |
+--------------+----------------+-----------------+------------+---------------+------------+--------------------+---------------------+
|            1 | Arun Kumar     | arun@example.com | 9876543210 | B.Tech        |          2 | Java, Python, MySQL | 2026-10-24 09:53:14 |
+--------------+----------------+-----------------+------------+---------------+------------+--------------------+---------------------+
```


MySQL 8.0 Command Line Cli
-> ap.application_status
-> FROM applicants a
-> JOIN applications ap
-> ON a.applicant_id = ap.applicant_id
-> JOIN jobs j
-> ON ap.job_id = j.job_id;
applicant_id	applicant_name	email	qualification	job_title	application_status
1	Arun Kumar	arun@example.com	B.Tech	Software Developer	Applied
2	Priya Sharma	priya@example.com	BCA	Web Developer	Shortlisted
3	Rahul Raj	rahul@example.com	B.Sc	Software Tester	Applied
4	Divya Nair	divya@example.com	MBA	HR Executive	Selected
5	Karthik S	karthik@example.com	MCA	Database Administrator	Interview Scheduled
5 rows in set (0.05 sec)
mysql> SELECT *
-> FROM applicants
-> WHERE applicant_name LIKE '%Arun%';
| applicant_id | applicant_name | email | phone | qualification | experience | skills | registration_date |
| 1 | Arun Kumar | arun@example.com | 9876543210 | B.Tech | 2 | Java, Python, MySQL | 2026-10-24 09:53:14 |
1 row in set (0.00 sec)
mysql>